Quantitative Integration Solutions

HN-FE01 | Instructor key | 100 points | 120 minutes

Award the stated subpart points for correct calculations with units and requested reasoning. Accept defensible alternatives in design questions. Numeric tolerance 0.05 unless an exact count is required. Keep this key separate from student sources.

Question 1 Resting expenditure models

HN-FE01-Q01 | 10 points

a) [4 points] 10(84)+6.25(178)-5(29)+5=1812.5 kcal/day.

b) [3 points] FFM=84(1-.25)=63 kg; 500+22(63)=1886; 370+21.6(63)=1730.8 kcal/day.

c) [3 points] They are equation outputs under supplied assumptions; no direct measurement/individual accuracy criterion is given.

Course modules 1, 2.

Question 2 Energy and macro allocation

HN-FE01-Q02 | 10 points

a) [4 points] 1812.5×1.6=2900; 2900×.85=2465 kcal/day.

b) [3 points] Protein 168 g=672 kcal; fat 63 g=567 kcal; carbohydrate(2465-1239)/4=306.5 g=1226 kcal; total 2465.

c) [3 points] Examples: supplied branch, inclusive multiplier, chosen adjustment/macro factors, estimated body composition. These are not measured individual needs.

Course modules 1, 2, 5.

Question 3 Activity accounting

HN-FE01-Q03 | 10 points

a) [4 points] 6×3.5×84/200×35=308.7 kcal.

b) [3 points] Incremental 257.25 kcal; reference 51.45 kcal.

c) [3 points] No; activity/reference components already included can be counted twice. State a non-overlapping component model before adding.

Course modules 1, 5.

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Question 4 Load estimates and session time

HN-FE01-Q04 | 10 points

a) [4 points] 90(1+6/30)=108 kg; model output, not a measured maximum.

b) [3 points] 81 kg unrounded; 80 kg nearest 2.5 kg total.

c) [3 points] 4×45+3×150=630 seconds=10 min 30 s; no rest after the last set.

Course modules 3, 5.

Question 5 Volume and progression audit

HN-FE01-Q05 | 15 points

a) [5 points] 4, 8, 12 respectively.

b) [5 points] Propose 92.5 kg total; 2.5/90×100=2.77778%; proposal only.

c) [5 points] Hold/review because not every current set reaches 12. Compound sets can contribute to multiple muscle categories; counts are not automatically physical-set totals.

Course modules 3, 5.

Question 6 External and internal moments

HN-FE01-Q06 | 15 points

a) [5 points] 120×.25=30 N·m; use perpendicular distance, not radius.

b) [5 points] 30/.03=1000 N under the explicitly simplified model.

c) [5 points] Real tasks have multiple muscles/forces, passive moments, varying internal leverage and dynamics; no direct measurement or injury model is supplied.

Course modules 4.

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Question 7 Trend and expenditure proxy

HN-FE01-Q07 | 15 points

a) [5 points] -.4 kg per week; 100(-.4)/84.2=-.475059%.

b) [5 points] 2465-7700(-.4/7)=2905 kcal/day; 5 kcal/day above the scenario.

c) [5 points] Both depend on assumptions/measurement inputs; fluid changes, intake error and static coefficient remain unresolved. Body mass is not identified fat mass.

Course modules 1, 2, 5.

Question 8 Repair an integrated report

HN-FE01-Q08 | 15 points

a) [5 points] Progression needs comparable complete evidence and the stated policy; included gross activity must not be added twice.

b) [5 points] 9, 10, 11, 11, 12, 6; total 59; week 6 reduction 50%. Proposed direct sets for one category.

c) [5 points] Scale decline is not exact fat loss; planned count is not measured MRV; export is a proposal snapshot. Check adherence/context and comparable performance before revising.

Course modules 1, 2, 3, 5.